PPL LAB 6

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Q1.Write a program in CUDA to add two vectors of length N using

a) block size as N

b) N threads

#include "cuda_runtime.h" #include <stdio.h>

__global__ void add(int *a, int *b, int *c,int n) { int i = blockDim.x*blockIdx.x + threadIdx.x; if(i < n) c[i] = a[i]+b[i]; }

int main() { int n; printf("enter size:\n"); scanf("%d",&n); int size = n * sizeof(int); int a[n],b[n],c[n]; int *d_a,*d_b,*d_c;

```
//allocate mem for a,b,c on device
cudaMalloc((void **)&d_a,size);
cudaMalloc((void **)&d_b,size);
cudaMalloc((void **)&d_c,size);

//input a,b
printf("enter %d values for a:\n",n);
for(int i = 0;i < n;i++)
    scanf("%d",&a[i]);
printf("enter %d values for b:\n",n);
for(int i = 0;i < n;i++)
    scanf("%d",&b[i]);
```

```
//copy values to device memory
cudaMemcpy(d_a,a,size,cudaMemcpyHostToDevice);
cudaMemcpy(d_b,b,size,cudaMemcpyHostToDevice);

add<<<n,1 >>>(d_a,d_b,d_c,n);
printf("c using %d blocks and 1 thread\n",n);
cudaMemcpy(c,d_c,size,cudaMemcpyDeviceToHost);
for(int i=0;i<n;i++)
    printf("%d ",c[i]);

printf("\nc using 1 block and %d threads\n",n);
add<<<1,n >>>(d_a,d_b,d_c,n);
cudaMemcpy(c,d_c,size,cudaMemcpyDeviceToHost);
for(int i=0;i<n;i++)
    printf("%d ",c[i]);
printf("\n");
cudaFree(d_a);
cudaFree(d_b);
cudaFree(d_c);
return 0;


}
```

Output:

```
STUDENT@MIT-ICT-LAB5-29:~/230905225/week5$ nvcc q1.cu
STUDENT@MIT-ICT-LAB5-29:~/230905225/week5$ ./a.out
enter size:
5
enter 5 values for a:
12 15 17 11 100
enter 5 values for b:
65 95 421 3 110
c using 5 blocks and 1 thread
77 110 438 14 210
c using 1 block and 5 threads
77 110 438 14 210
```

Q2. Implement a CUDA program to add two vectors of length N by keeping the number of threads per block as 256(constant) and vary the number of blocks to handle N elements.

#include "cuda_runtime.h"

#include <stdio.h>

#include <math.h>

__global__ void add(int *a,int *b,int *c,int n){ int i = blockDim.x*blockIdx.x + threadIdx.x; if(i < n) c[i] = a[i]+b[i]; }

int main(){ int n; printf("enter size:\n"); scanf("%d",&n); int size = n * sizeof(int); int a[n],b[n],c[n]; int *d_a,*d_b,*d_c;

```
cudaMalloc((void**)&d_a,size);
cudaMalloc((void**)&d_b,size);
cudaMalloc((void**)&d_c,size);

printf("enter %d elements for a\n",n);
for(int i = 0;i < n;i++)
    scanf("%d",&a[i]);

printf("enter %d elements for b\n",n);
for(int i = 0;i < n;i++)
    scanf("%d",&b[i]);

cudaMemcpy(d_a,a,size,cudaMemcpyHostToDevice);
cudaMemcpy(d_b,b,size,cudaMemcpyHostToDevice);
```

```
add<<<ceil(n/256.0),256>>>(d_a,d_b,d_c,n);

cudaMemcpy(c,d_c,size,cudaMemcpyDeviceToHost);
printf("result\n");
for(int i = 0;i< n;i++)
    printf("%d ",c[i]);
printf("\n");
cudaFree(d_a);
cudaFree(d_b);
cudaFree(d_c);
return 0;


}
```

Output:

```
STUDENT@MIT-ICT-LAB5-29:~/230905225/week5$ nvcc q2.cu
STUDENT@MIT-ICT-LAB5-29:~/230905225/week5$ ./a.out
enter size:
6
enter 6 elements for a
11 25 48 986 301 254
enter 6 elements for b
98 5 210 325 655
4
result
109 30 258 1311 956 258
```

Q3. Write a program in CUDA to process a 1D array containing angles in radians to generate sine of the angles in the output array. Use appropriate function.

```
#include "cuda_runtime.h"

 #include <stdio.h>

__global__ void convertToSine(float *angles,float sines,int n){ int i =
blockDim.xblockIdx.x+threadIdx.x; if(i<n) sines[i]=sinf(angles[i]); }
```

int main(){ int n; printf("enter n\n"); scanf("%d",&n); float angles[n],sines[n]; float *d_angles,*d_sines; int size = n * sizeof(float);

```
cudaMalloc((void**)&d_sines,size);
cudaMalloc((void**)&d_angles,size);

printf("enter %d angles in radians\n",n);
for(int i = 0;i < n;i++)
    scanf("%f",&angles[i]);

cudaMemcpy(d_angles,angles,size,cudaMemcpyHostToDevice);

convertToSine<<<1,n>>>(d_angles,d_sines,n);
cudaMemcpy(sines,d_sines,size,cudaMemcpyDeviceToHost);
printf("sines of the angles\n");
for(int i = 0;i < n;i++)
    printf("%f ",sines[i]);
printf("\n");
cudaFree(d_angles);
cudaFree(d_sines);
return 0;

}
```

Output:

```
STUDENT@MIT-ICT-LAB5-29:~/230905225/week5$ nvcc q3.cu
STUDENT@MIT-ICT-LAB5-29:~/230905225/week5$ ./a.out
enter n
3
enter 3 angles in radians
3.14 1.57 0
sines of the angles
0.001593 1.000000 0.000000
STUDENT@MIT-ICT-LAB5-29:~/230905225/week5$
```